Letter of Recommendation – Arjit Yadav

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Letter of Recommendation for Mr. Arjit Yadav

To whom it may concern,

It is my pleasure to recommend Mr. Arjit Yadav, who completed his M.Sc. in Computer Science at the University of Stuttgart. I supervised his Master's thesis and continue to work closely with him on a joint research publication. Based on this collaboration, I can speak with confidence about his technical ability, his independence and the quality of his work.

In his Master's thesis, *Energy Efficient Semantic Segmentation Using Spiking Neural Networks*, Mr. Yadav investigated how spiking neural networks can perform dense, pixel-wise prediction tasks at a fraction of the energy cost of conventional deep networks. This topic requires a solid grasp of deep learning for computer vision as well as of the specific training methods and constraints of spiking models. He mastered both areas quickly and delivered a thesis with a very good grading recommendation from me.

After his thesis, Mr. Yadav continues working with our group on a paper on *Energy-Efficient Semantic Segmentation on SpiNNaker2*. In this project he is responsible for implementing a spiking neural network for semantic segmentation. Beyond the model itself, he extended two existing python frameworks (py-spinnaker2, NIR) with support for up-convolution, the layer type needed to restore spatial resolution in segmentation networks. This contribution is essential for deploying the model on the SpiNNaker2 neuromorphic hardware. It required him to work his way into an existing research codebase and extend it cleanly, which he did with notable ease.

Across both projects, Mr. Yadav has shown fast and reliable coding skills. He turns ideas into working, well-structured implementations quickly, and his code can be trusted. He identifies problems on his own, proposes solutions and drives his tasks forward without needing close guidance. At the same time, he takes great care in documenting his work and reporting results. His reports are clear and thorough, which made our discussions efficient and makes his work easy for others to build on.

In summary, Mr. Yadav is a capable, independent and conscientious engineer with a strong background in spiking neural networks, computer vision and neuromorphic computing. I recommend him highly and without reservation for any position. I am confident he will be a valuable addition to your team. Please do not hesitate to contact me if you need any further information.

Sincerely,

